VCF Private Cloud Deployment – Network Architecture Overview

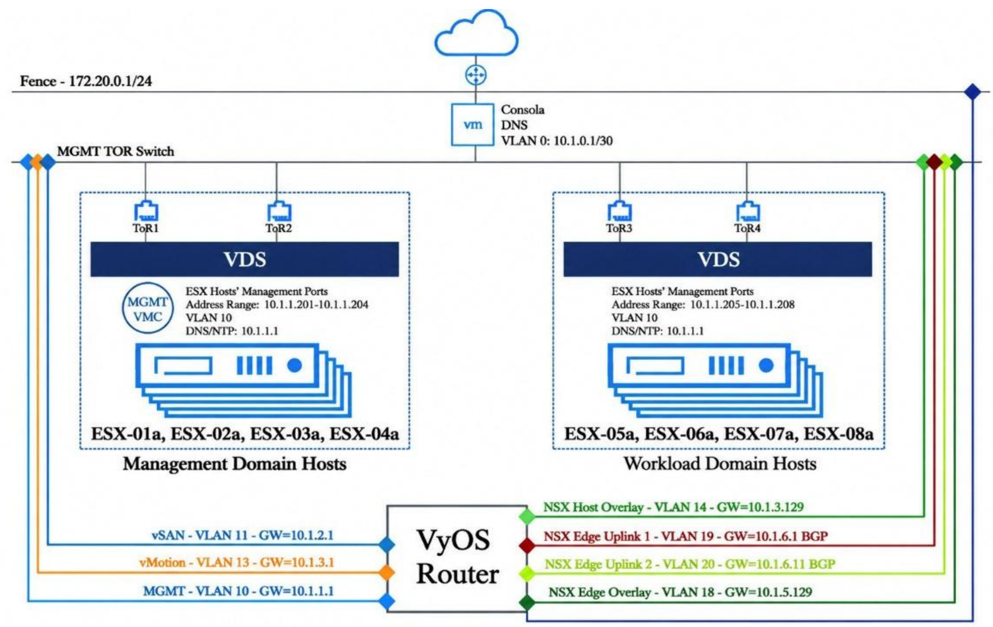
The VCF Private Cloud environment is built on a **redundant network** architecture consisting of management and workload domains connected through a pair of Top-of-Rack (ToR) switches and a VyOS router.

The **Management Domain** includes four ESXi hosts (ESX-01a to ESX-04a), while the **Workload Domain** consists of four additional ESXi hosts (ESX-05a to ESX-08a). Both domains use dedicated Virtual Distributed Switches (VDS).

The network is segmented into dedicated VLANs for **Management, vSAN, vMotion, and NSX overlay/edge traffic**. A VyOS router provides the required gateways and routing between these networks, including BGP connectivity for the NSX Edge uplinks.

A dedicated **DNS/Console VM** provides DNS and management access for the environment. The management network uses VLAN 10 (10.1.1.0/24), while vSAN and vMotion use VLANs 11 and 13, respectively. NSX traffic is isolated across dedicated overlay and Edge VLANs.

This architecture provides the **connectivity, network segmentation, redundancy, and routing foundation required for the VCF Private Cloud deployment.**

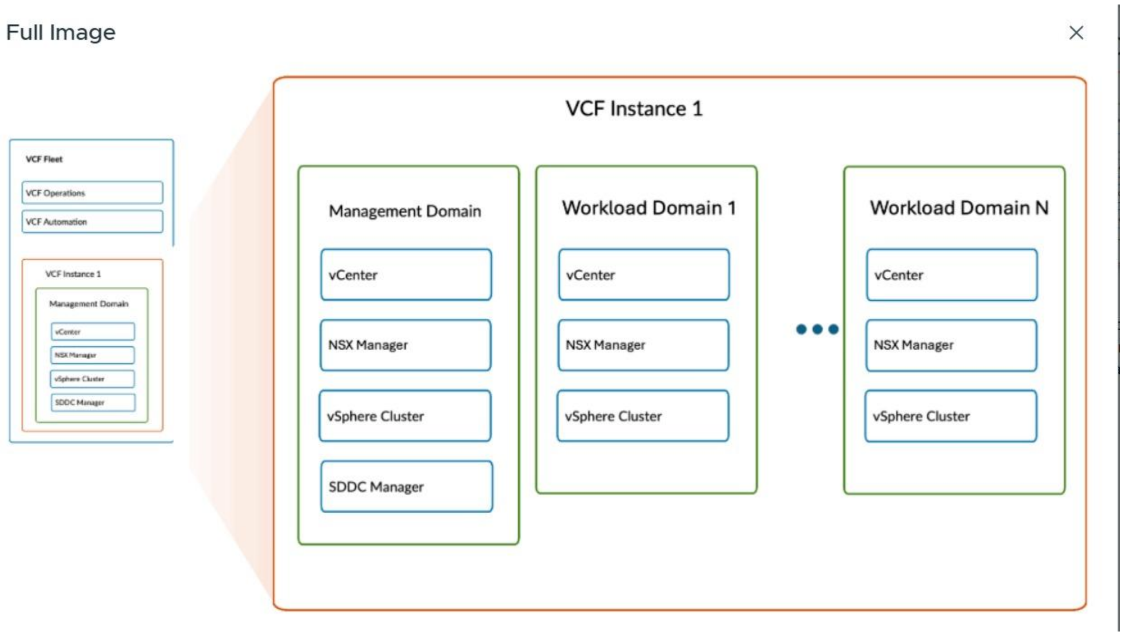


VCF Private Cloud Logical Architecture

The **VCF Private Cloud** is organized into a **Management Domain** and one or more **Workload Domains** within a VCF Instance.

The **Management Domain** hosts the core VCF infrastructure, including **vCenter, NSX Manager, the vSphere Cluster, and SDDC Manager**. Workload Domains provide dedicated environments for running tenant and application workloads, with each domain containing its own **vCenter, NSX Manager, and vSphere Cluster**.

At the platform level, **VCF Fleet** provides centralized management capabilities through **VCF Operations and VCF Automation**, enabling consistent operations and management across the VCF environment.



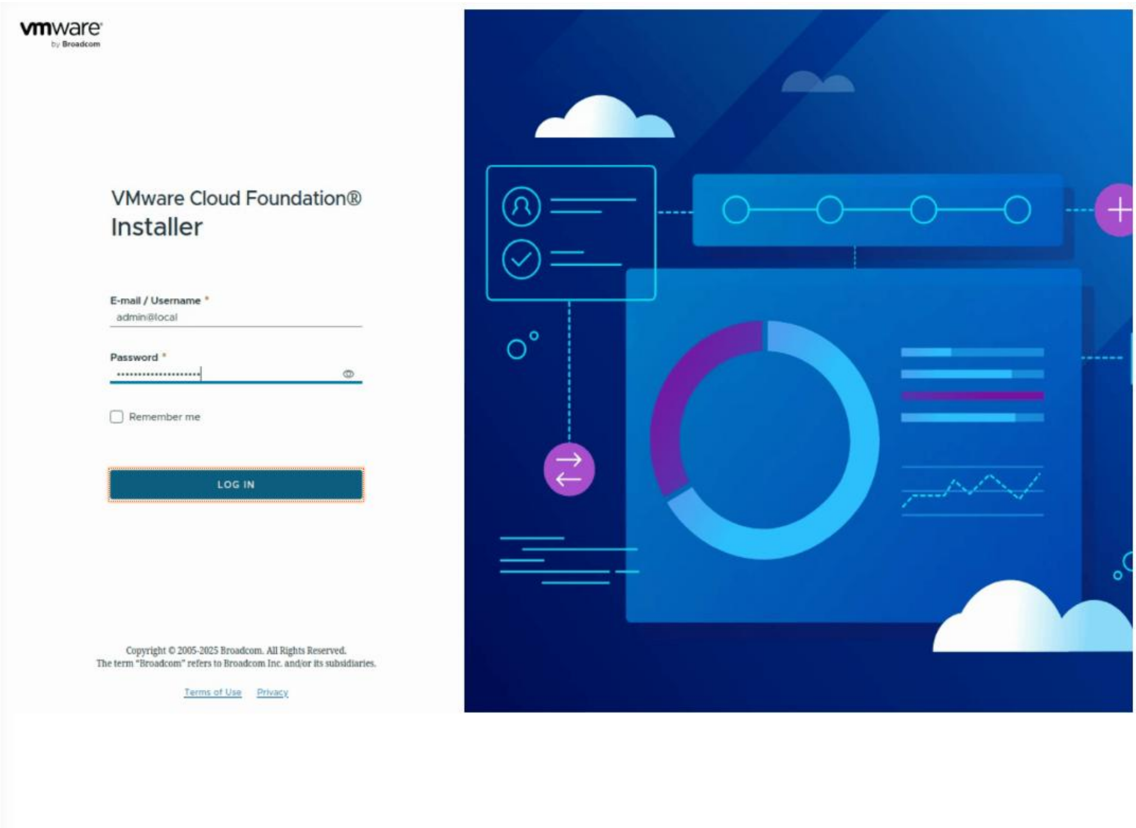
VMware Cloud Foundation Installer

Initial Login

After deploying the VMware Cloud Foundation (VCF) Installer virtual appliance, navigate to the IP address of the VCF Installer appliance from a browser. The default username of 'admin@local' for initial login should auto populate.

To login:

1. Click to enter the password.
2. Click on the '**LOG IN**' button.

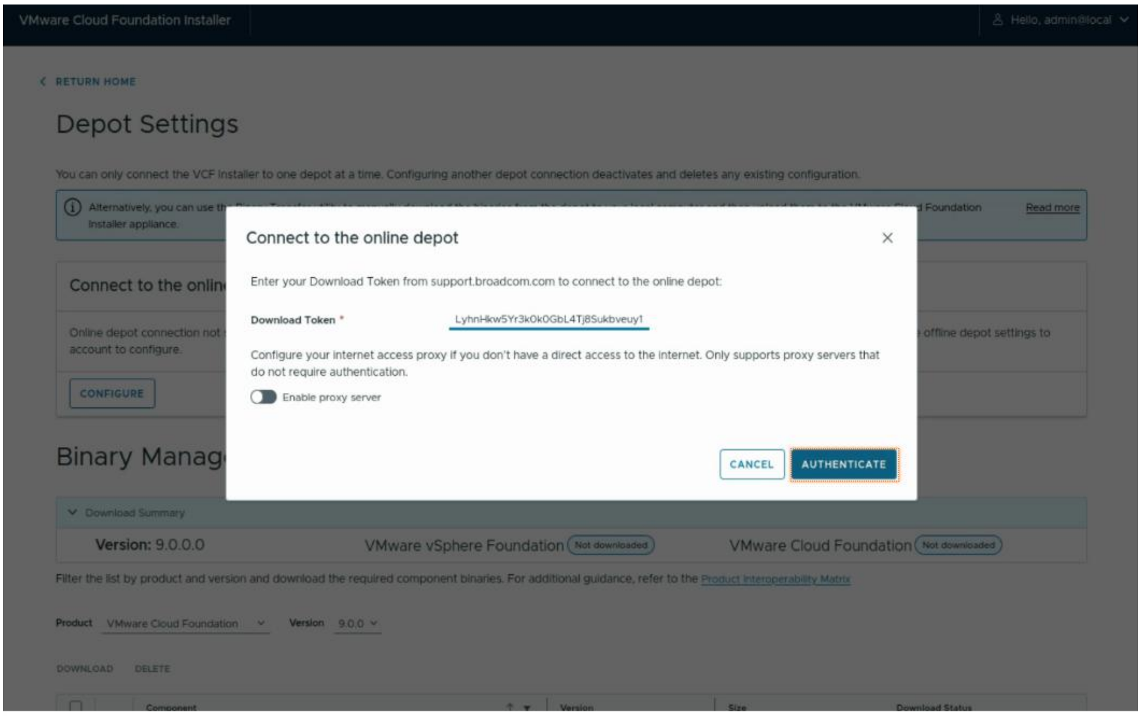


Configure Online Depot Connection

The first step upon login is to set up the VCF depot connection and download the required software binaries. VCF Installer appliance ships without software binaries. You will need your Broadcom download token that you can obtain from support.broadcom.com.

1. Click '**DEPOT SETTINGS AND BINARY MANAGEMENT**'.
2. Click '**CONFIGURE'** to enter your Broadcom entitlement token.
3. To enter your download token, click the Download Token text field. For the purpose of this demonstration, the token will automatically be entered.
4. Click '**AUTHENTICATE**' to verify and save your token.

The online depot connection is now active.

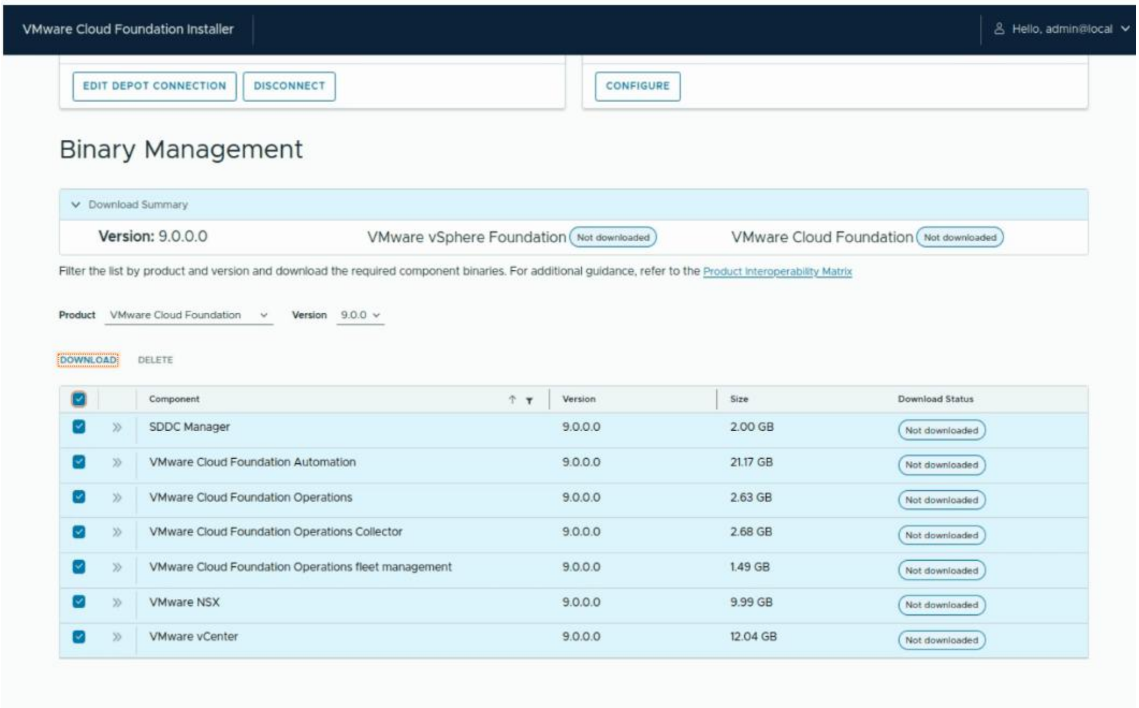


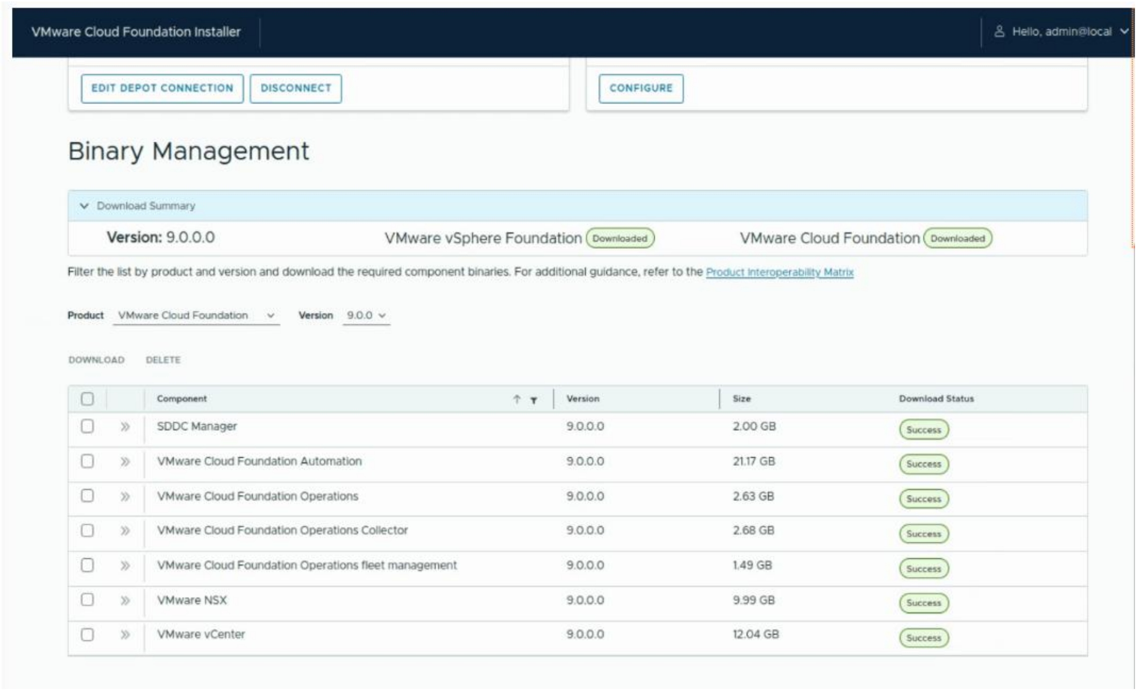
Download Software Binaries

We will now download the required software binaries to install VCF.

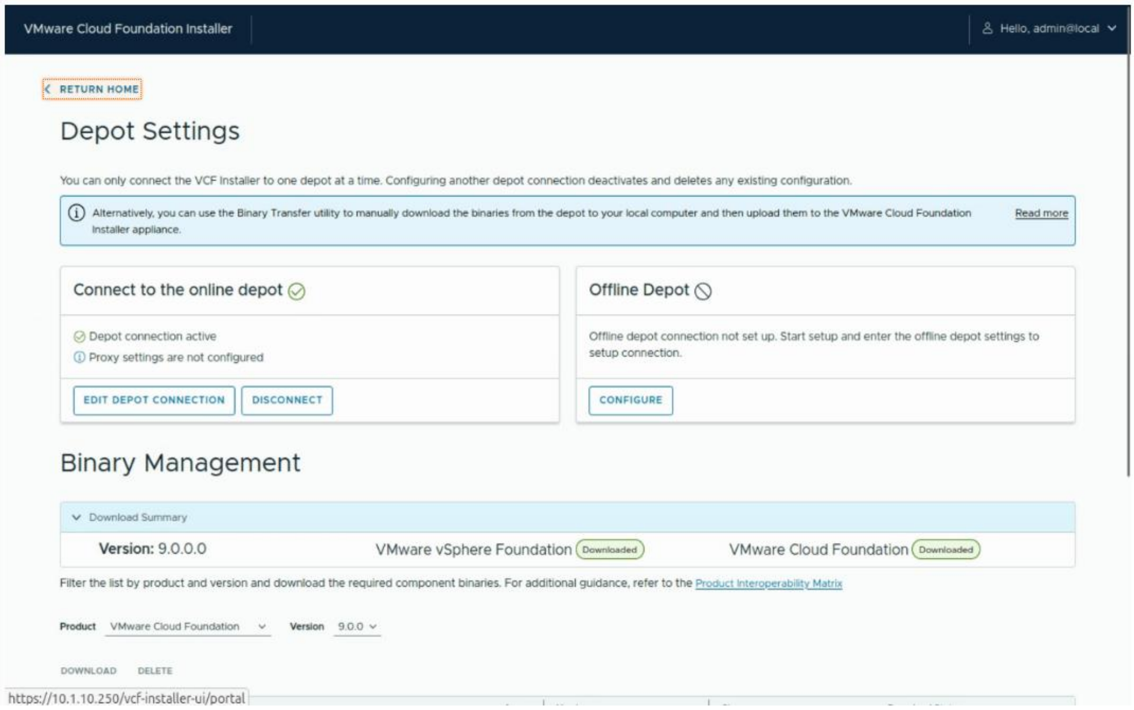
1. Click on the vertical scroll bar to the right to scroll down.
2. We will be downloading all the required software for the pre-selected VCF Version 9.0.0. Click on the checkbox on the top left of the table to select all.
3. Click '**DOWNLOAD**' to start downloading from the configured online depot completed in previous steps.

The required software binaries will be downloaded, validated and show a status of 'Success' when complete.





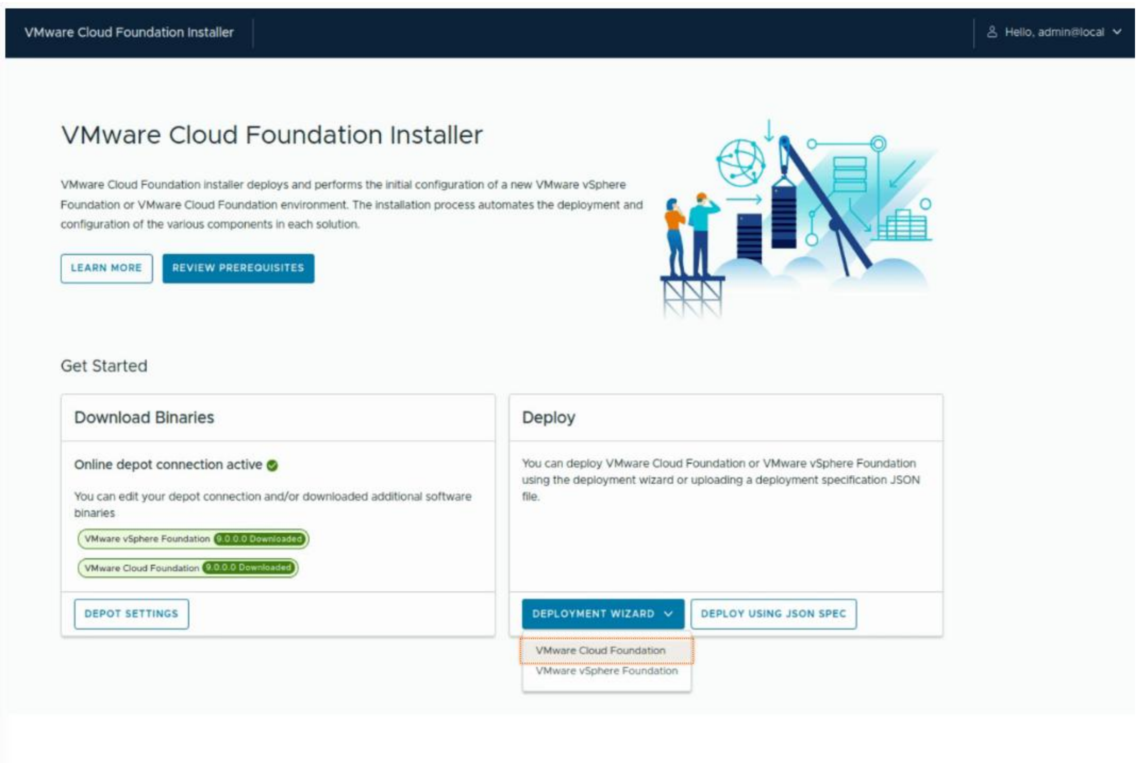
1. Scroll up the window by clicking on the top right scroll bar.
2. Click '**RETURN HOME**' to return to the main screen for the next step.



Deploy VMware Cloud Foundation

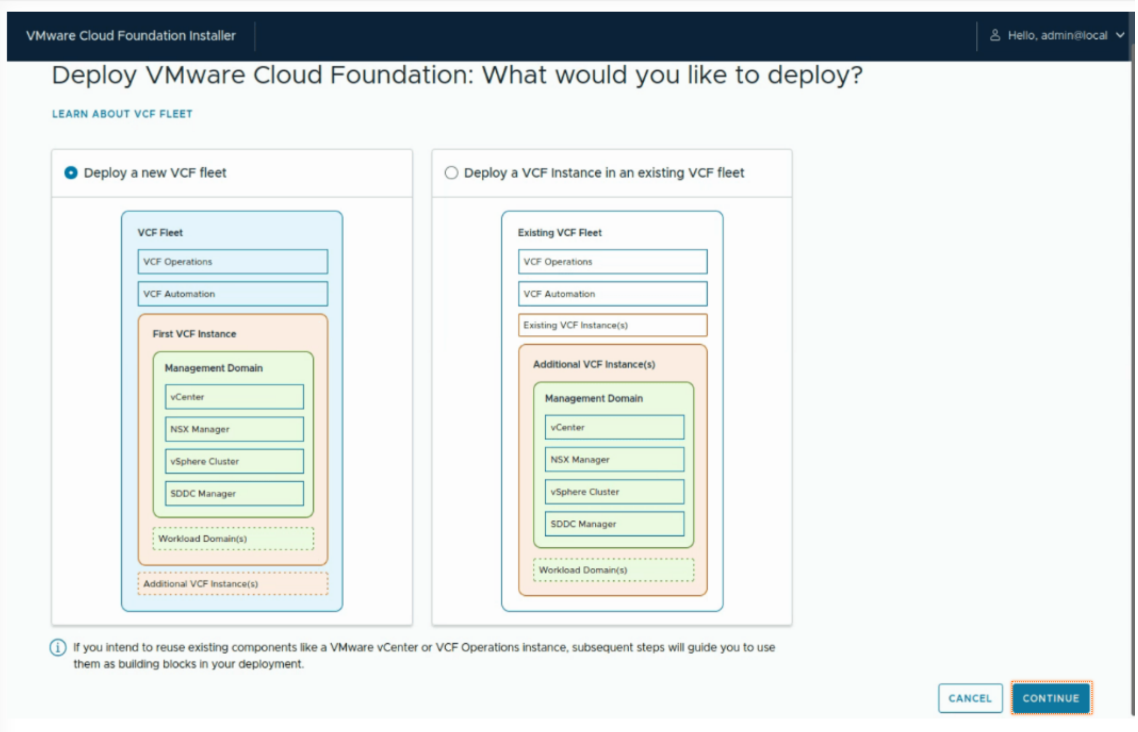
We will now begin the deployment of VMware Cloud Foundation.

1. Click '**DEPLOYMENT WIZARD**' to start the wizard.
2. Select '**VMware Cloud Foundation**' from the drop down menu as this is the cloud platform we will be deploying.



In this simulator, we will be deploying a new VCF Fleet. This option has been pre-selected. The VCF Installer can be used to deploy a VCF Instance in an existing VCF fleet as well.

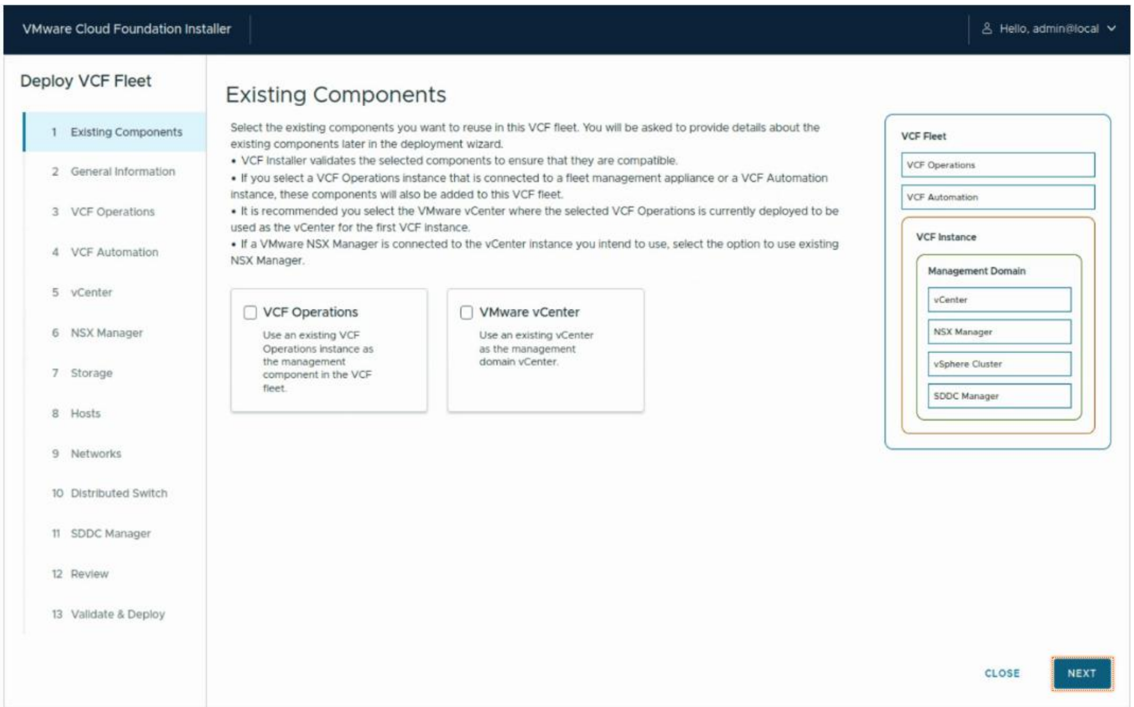
1. Click on the vertical scroll bar on the bottom right to scroll down.
2. Click '**CONTINUE**' to begin.



Existing Components

If you have existing VCF Operations or VMware vCenter already deployed, you can reuse them for this VCF Fleet deployment. In this simulator, we will be deploying new.

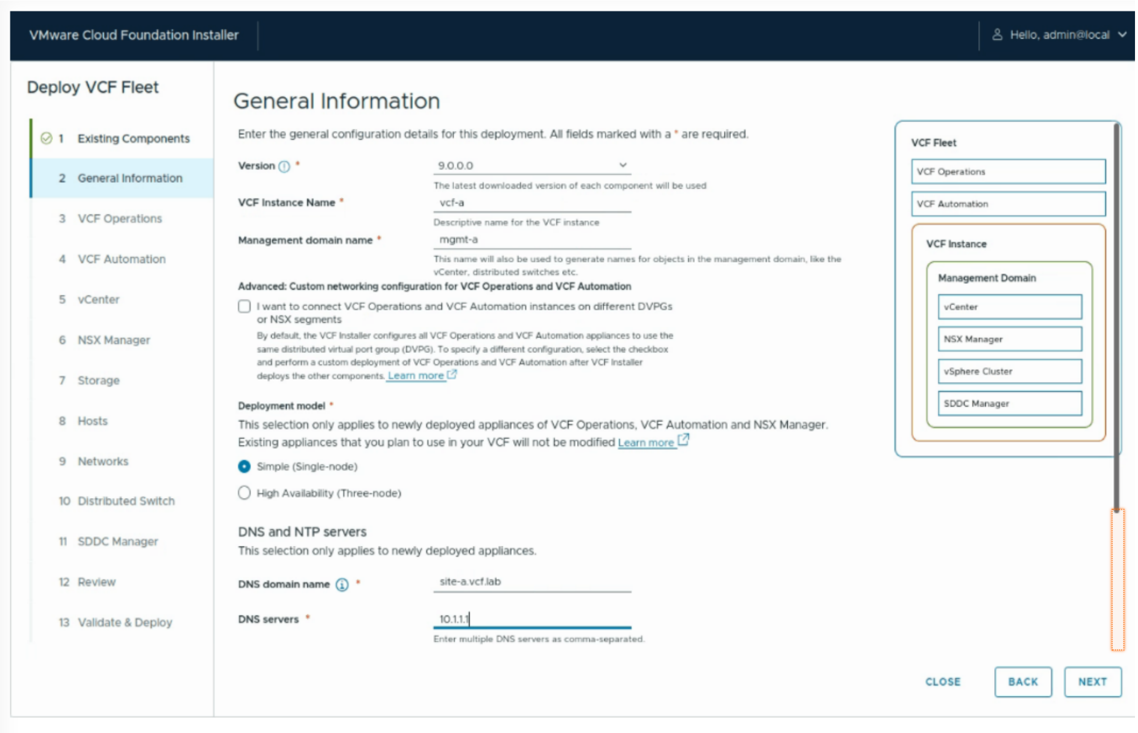
1. Click '**NEXT**' to proceed to the next step.



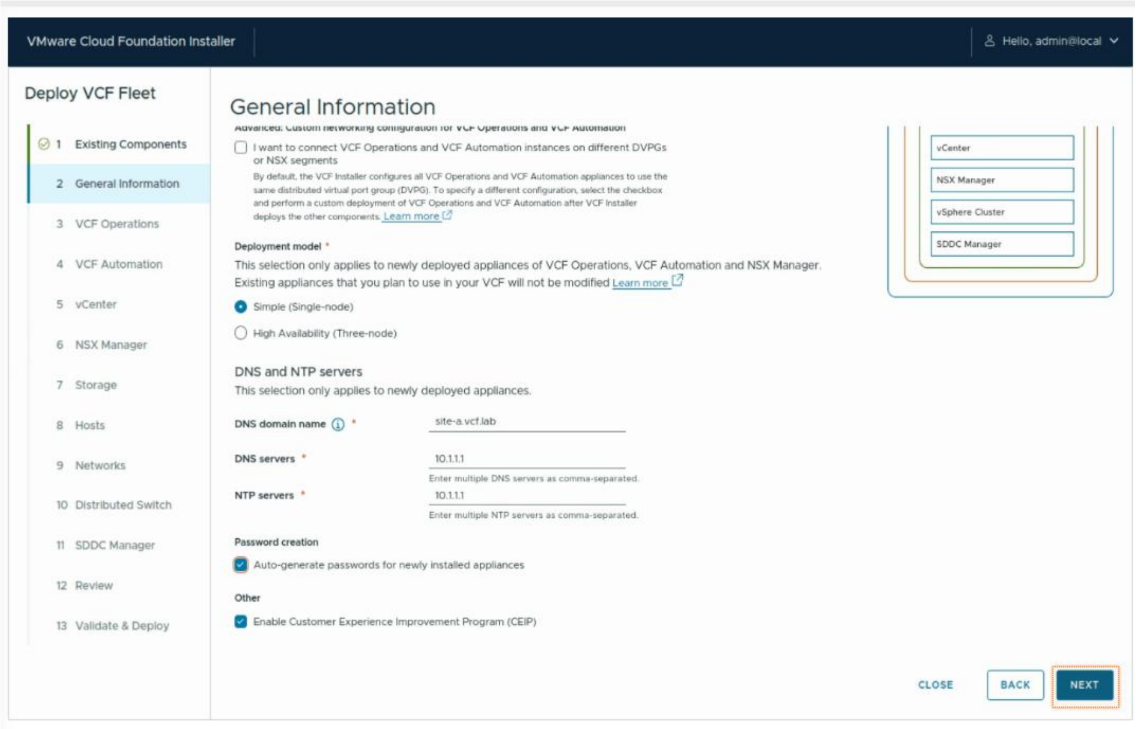
General Information

We will begin filling out general information about this VCF deployment. The default and only version of VCF 9.0.0 has been preselected.

1. Click on the **VCF Instance Name** text field.
2. Type the name '**vcf-a**'.
3. Click on the **Management domain name** text field.
4. Type the name '**mgmt-a**'.
5. You have the option to choose between a single-node or three-node deployment model for the various VCF appliances. For this simulation, select '**Simple (Single-node)**'.
6. Click on the **DNS domain name** text field.
7. Type **'site-a.vcf.lab**'.
8. **Click on the DNS servers** text field.
9. Type '**10.1.1.1**' for this lab.



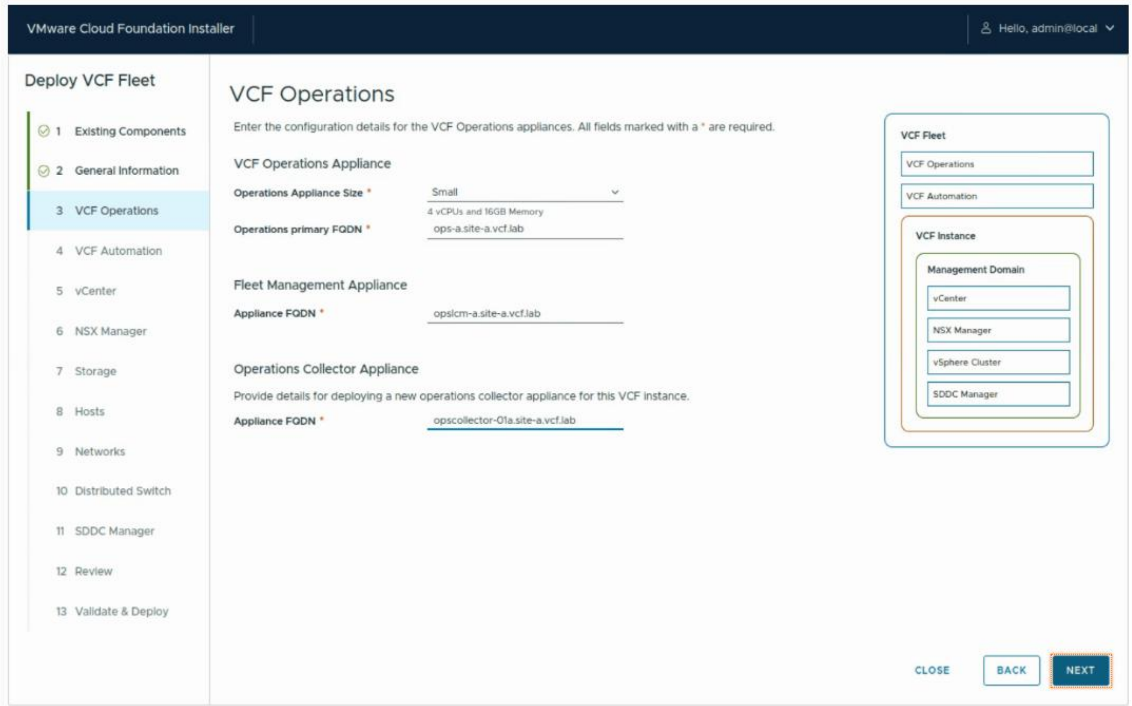
10. Click on the vertical scroll bar to the right to scroll down to view more.
11. Click on the '**NTP servers**' text field.
12. Enter '**10.1.1.1**' for this lab.
13. You have the option to specify your own passwords for the newly installed appliances or let the system generate.
    For this simulation, check the box '**Auto-generate passwords for newly installed appliances**'.
14. Click '**NEXT**' to proceed to the next section.



VCF Operations

This section asks for information related to the VCF Operations appliance deployment. As we had specified a 'Simple (Single-Node)' deployment model previously, only information for a single-node deployment is requested.

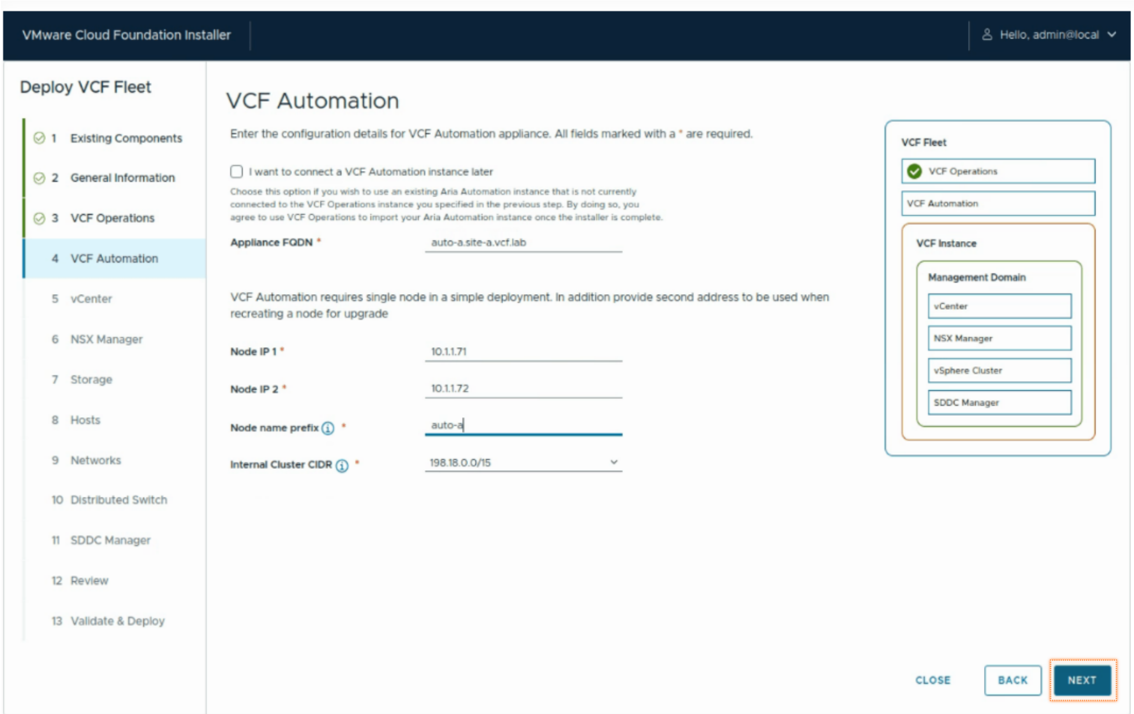
1. Click on the drop-down list to choose the **Operations Appliance Size**.
2. You have a choice between Extra Small, Small, Medium, Large and Extra Large. Click on '**Small**'.
3. Click on the **Operations primary FQDN** text field.
4. Type '**ops-a.site-a.vcf.lab**'.
5. Click on the Fleet Management Appliance **Appliance FQDN** text field.
   The Fleet Management Appliance is an appliance that is deployed to manage deployments and lifecycle management of VCF Fleet components.
6. Type '**opslcm-a.site-a.vcf.lab**'.
7. Click on the Operations Collector Appliance **Appliance FQDN** text field.
8. Enter '**opscollector-01a.site-a.vcf.lab**'.
9. Click '**NEXT**' to proceed to the next section.



VCF Automation

This section asks for information related to the VCF Automation appliance deployment. As we had specified a 'Simple (Single-Node)' deployment model previously, only information for a single-node deployment is requested. If you have an existing Aria Automation instance and wish to use it instead of deploying new, you can select the checkbox 'I want to connect a VCF Automation instance later'. For this simulation, we will deploy a new appliance.

1. Click on the **Appliance FQDN** text field.
2. Type '**auto-a.site-a.vcf.lab**'.
3. Click on the **Node IP 1** text field.
4. Type '**10.1.1.71**'.
5. Click on the **Node IP 2** text field.
6. Type '**10.1.1.72**'.
7. Click on the **Node name prefix** text field.
8. Type '**auto-a**'.
9. Click '**NEXT**' to proceed to the next section.



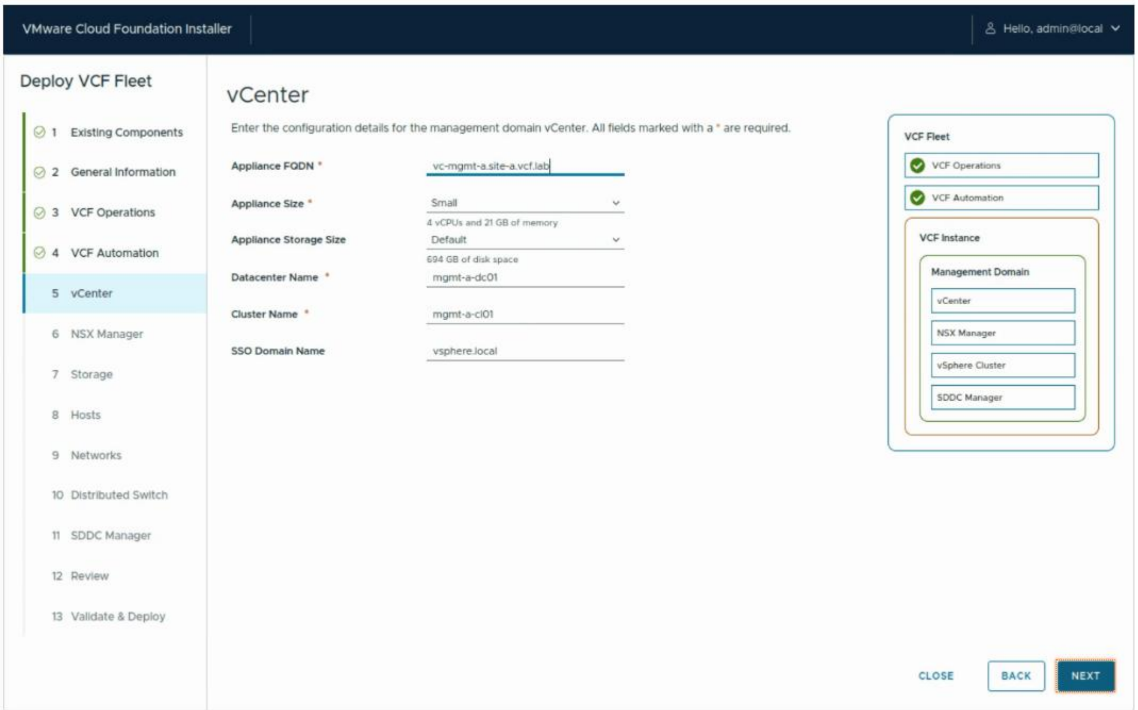
vCenter

This section asks for information related to the vCenter appliance deployment. This will be the first vCenter appliance deployed for the management domain.

1. Click on the **Appliance FQDN** text field.
2. Type '**vc-mgmt-a.site-a.vcf.lab**'.

We will leave all the other fields as the auto-generated defaults. Take a moment to observe the customizable fields required.

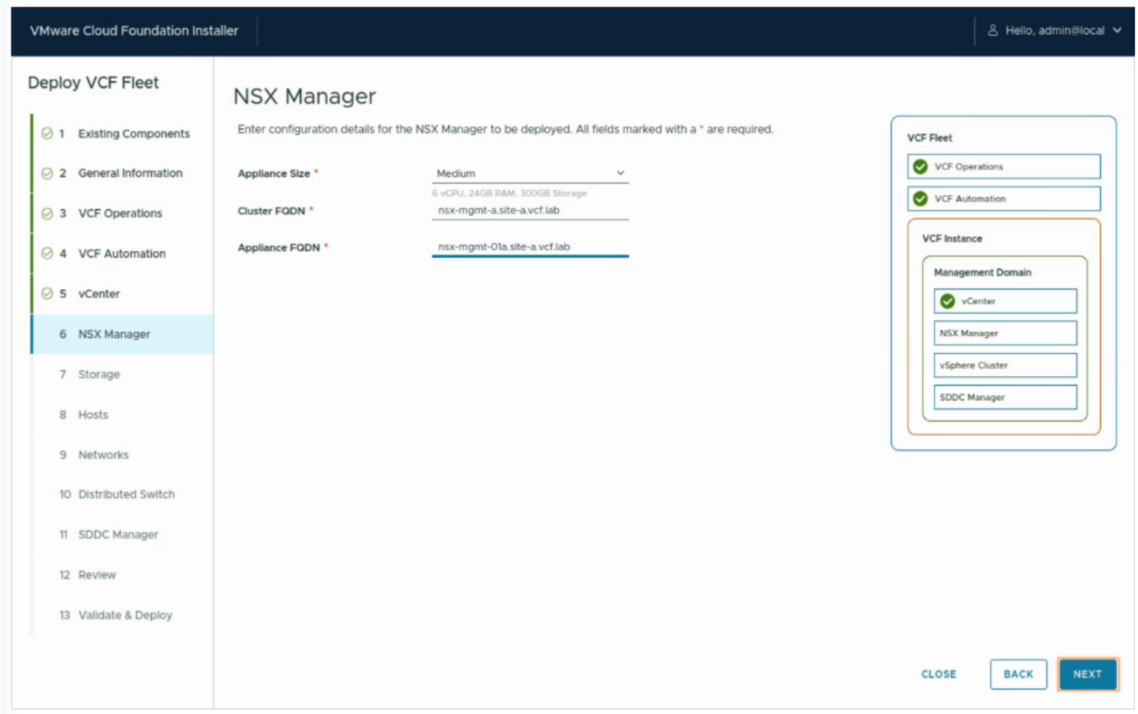
1. Click '**NEXT**' to proceed to the next section.



NSX Manager

This section asks for information related to the NSX Manager appliance deployment. As we had specified a 'Simple (Single-Node)' deployment model previously, only information for a single-node deployment is requested. You have a choice of NSX Manager appliance sizes to choose from. We will choose the pre-selected default size of Medium.

1. Click on the **Cluster FQDN** text field.
2. Type '**nsx-mgmt-a.site-a.vcf.lab**'.
3. Click on the **Appliance FQDN** text field.
4. Type '**nsx-mgmt-01a.site-a.vcf.lab**'.
5. Click '**NEXT**' to proceed to the next section.



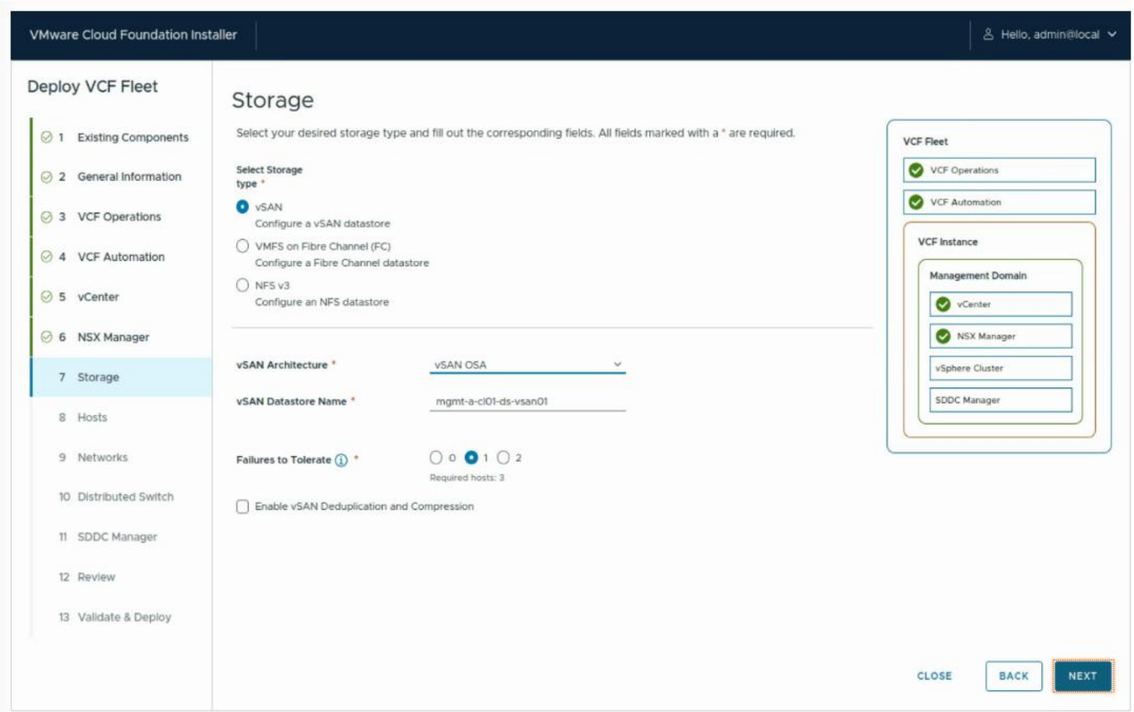
Storage

This section asks for information related to principal storage that will be used for the management domain. You have a choice of vSAN, VMFS on Fibre Channel (FC) or NFS v3.

1. Click on the **vSAN Architecture** drop down list.
2. You have a choice of vSAN ESA or vSAN OSA. For this simulation, click on '**vSAN OSA**'.

We will leave all the other options with auto-generated defaults. Take a moment to observe the customizable options available.

1. Click '**NEXT**' to proceed to the next section.

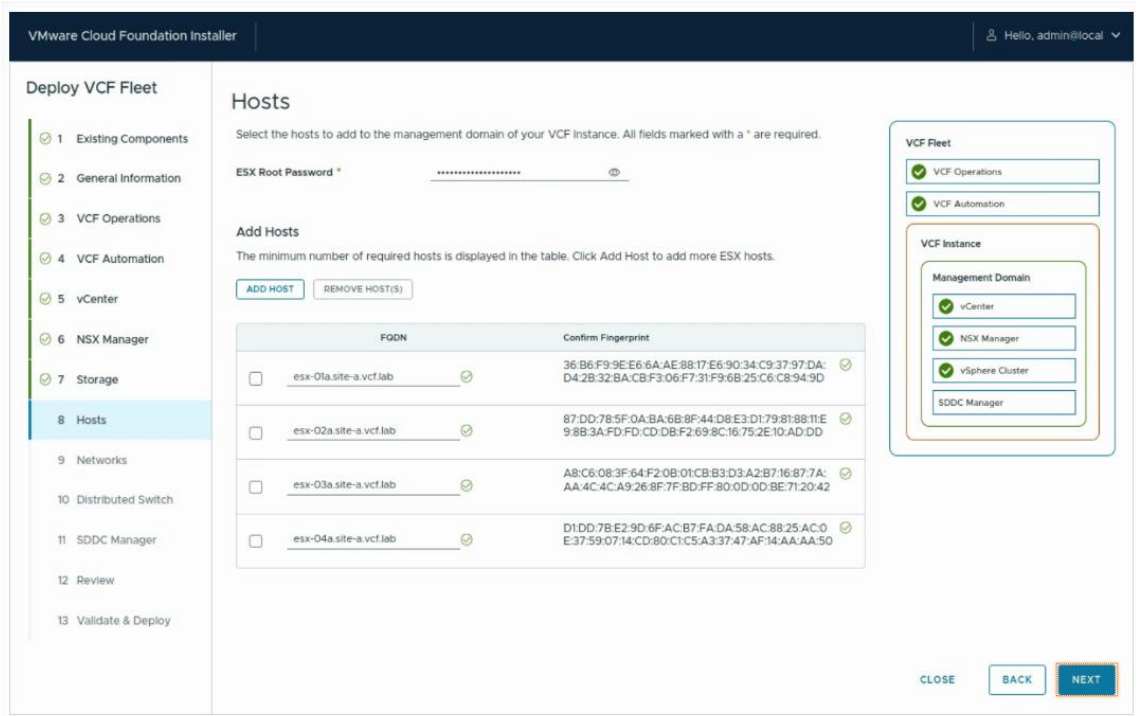


Hosts

This section asks for information related to ESX hosts that will be used for the management domain. For this simulation, we will be deploying with four hosts.

1. Click on the **ESX Root Password** text field to enter the password.
2. Click '**ADD HOST**' to add a fourth ESX host entry.
3. Click on the '**enter FQDN**' text field of the first host.
4. Type '**esx-01a.site-a.vcf.lab**'.
5. Click '**ADD**' to add the host.
6. Click on the '**enter FQDN**' text field of the second host.
7. Type '**esx-02a.site-a.vcf.lab**'.
8. Click '**ADD**' to add the host.
9. Click on the '**enter FQDN**' text field of the third host.
10. Type '**esx-03a.site-a.vcf.lab**'.
11. Click '**ADD**' to add the host.
12. Click on the '**enter FQDN**' text field of the fourth host.
13. Type '**esx-04a.site-a.vcf.lab**'.
14. Click '**ADD**' to add the host.
15. Click '**CONFIRM ALL FINGERPRINTS**' to confirm the SSL certificate fingerprint of all hosts.

16. Click '**NEXT**' to proceed to the next section.



Networks

This section asks for information related to networks that will be used by the ESX hosts in the management domain. We will need the network information for ESX Management, VM Management, vMotion and vSAN. For this simulation, we will be using the same network for both ESX and VM Management.

ESX Management Network

1. Click on the **VLAN ID** text field.
2. Type '**10**'.
3. Click on the **CIDR Notation** text field.
4. Type '**10.1.1.0/24**'.
5. Click on the **Gateway** text field.
6. Type '**10.1.1.1**'.

VM Management Network

1. Select the check box **Use same inputs from ESX Management Network**.

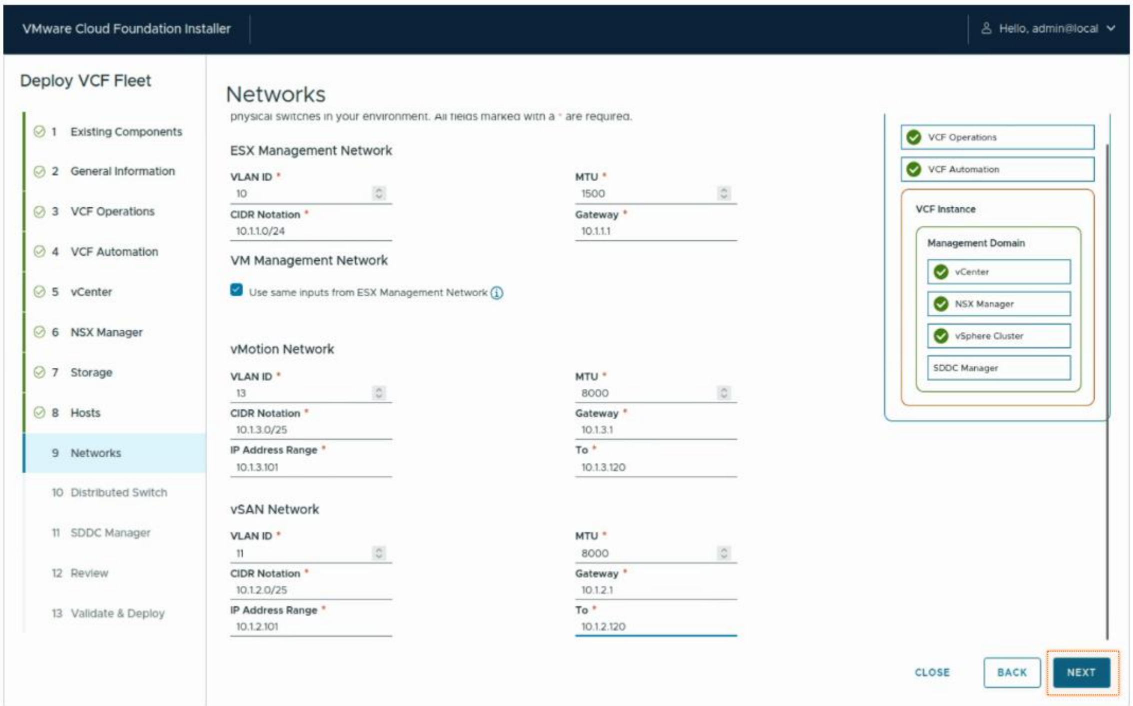
vMotion Network

1. Click on the **VLAN ID** text field.
2. Type '**13**'.
3. Click on the **CIDR Notation** text field.
4. Type '**10.1.3.0/25**'.

5. Click on the **MTU** text field.
6. Delete and replace the text with **8000**.
7. Click on the **Gateway** text field.
8. Type '**10.1.3.1**'.
9. Click on the **IP Address Range** text field.
10. Type '**10.1.3.101**'.
11. Click on the 'T**o'** text field.
12. Type '**10.1.3.120**'.

vSAN Network

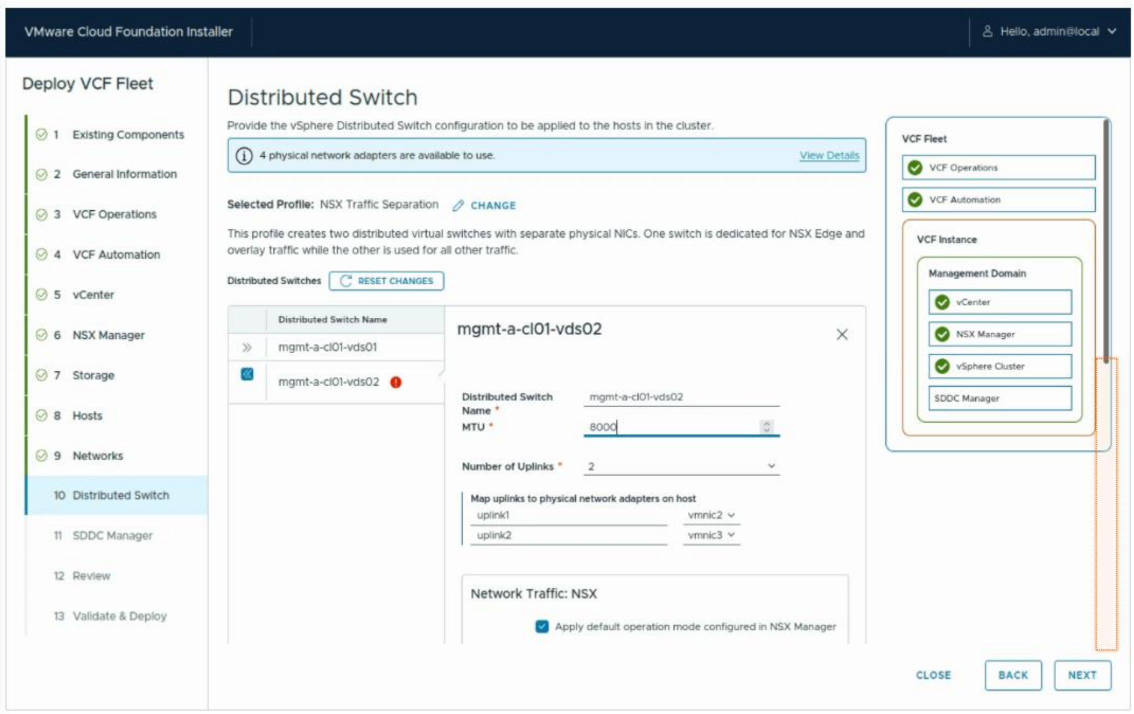
1. Click on the **VLAN ID** text field.
2. Type '**11**'.
3. Click on the **MTU** text field.
4. Delete and replace the text with **8000**.
5. Click on the **CIDR Notation** text field.
6. Type '**10.1.2.0/25**'.
7. Click on the **Gateway** text field.
8. Type '**10.1.2.1**'.
9. Click on the vertical scroll bar to the right to scroll down the page.
10. Click on the **IP Address Range** text field.
11. Type '**10.1.2.101**'.
12. Click on the 'T**o'** text field.
13. Type '**10.1.2.120**'.
14. Click '**NEXT**' to proceed to the next section.



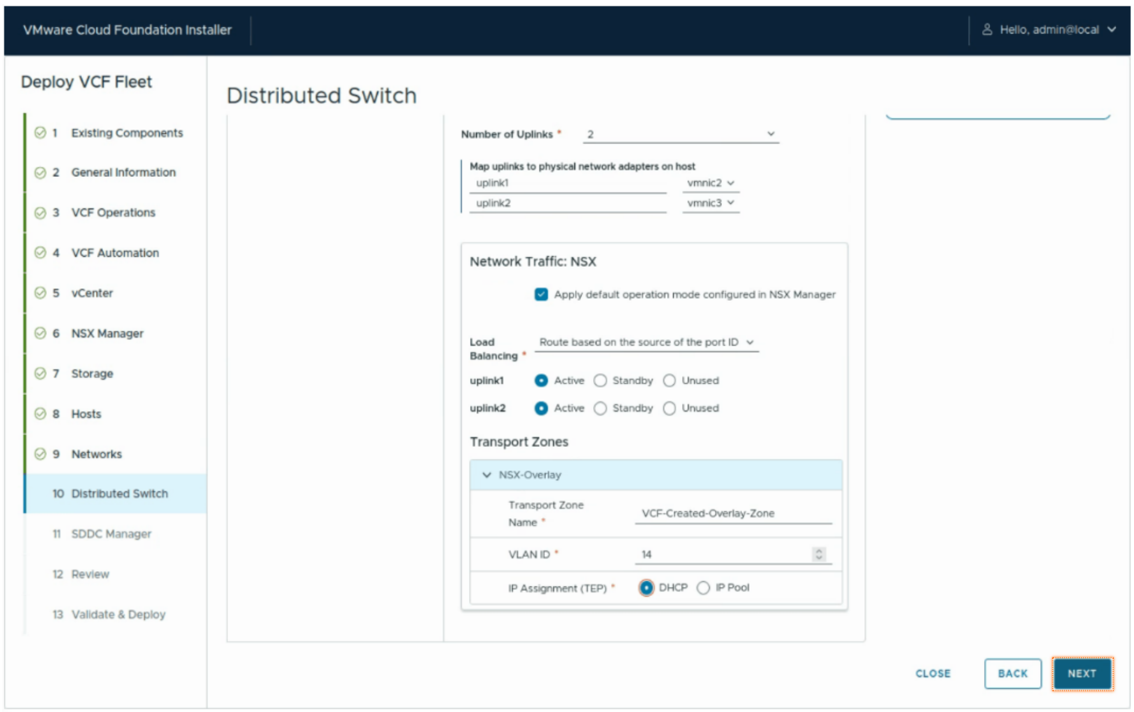
Distributed Switch

This section asks for information related to the distributed switch configuration that will be used by the management domain cluster. For this simulation, we have 4 pNics for each ESX host and we will configure them for NSX Traffic Separation where the NSX traffic will have its own dedicated pair of uplinks.

1. Click on the vertical scroll bar to the right to scroll down the page.
2. Click '**SELECT**' on the **NSX Traffic Separation** box to create 2 distributed switches. One for the NSX traffic and another for others.
3. We are now presented with configuration options for the distributed switches. Click on the **MTU** text field of **mgmt-a-cl01-vds02**.
4. Delete and replace the text with **8000**.



5. Click on the vertical scroll bar to the right to scroll down the page.
6. Click on the **NSX-Overlay VLAN ID** text field.
7. Type '**14**'.
8. Select '**DHCP**' for the **IP Assignment (TEP)** option.
9. Click '**NEXT**' to proceed to the next section.



SDDC Manager

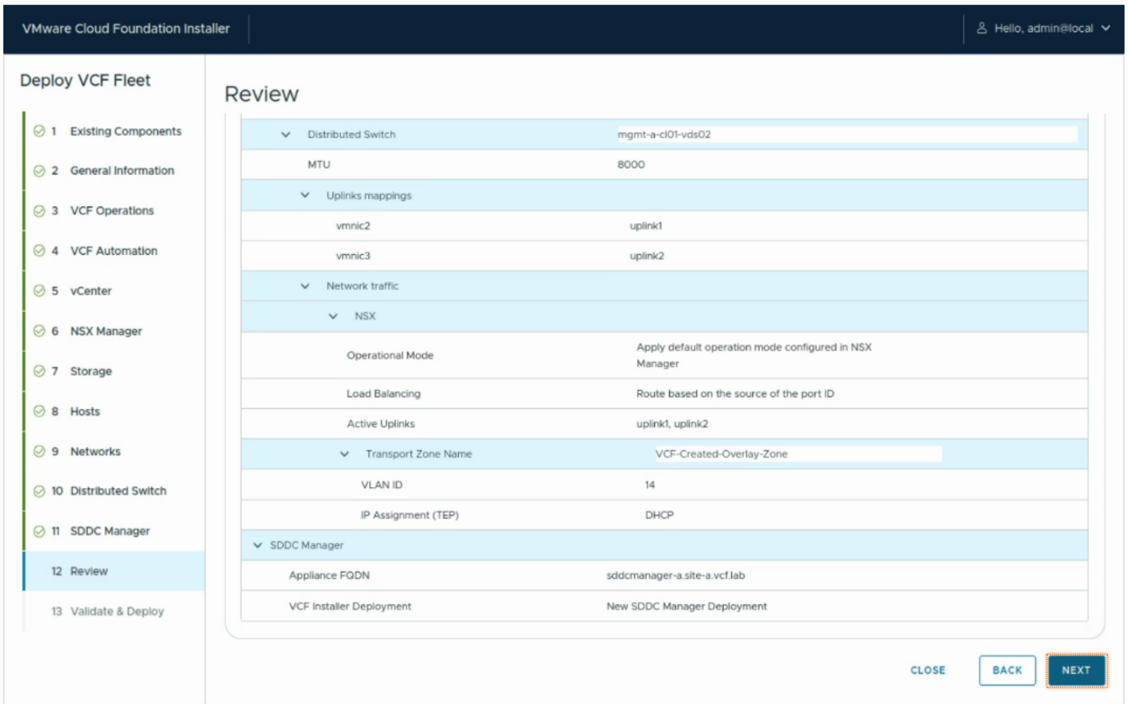
This section asks for information related to the SDDC Manager appliance deployment.

1. Click on the **Appliance FQDN** text field.
2. Type '**sddcmanager-a.site-a.vcf.lab**'.
3. Click '**NEXT**' to proceed to the next section.

Review

We are now ready to review the information entered before we proceed to validate and deploy VCF. Take a moment to review the information entered.

1. Click on the vertical scroll bar to the right to scroll down the page to see more.
2. When ready, click '**NEXT**' to proceed to the next section.

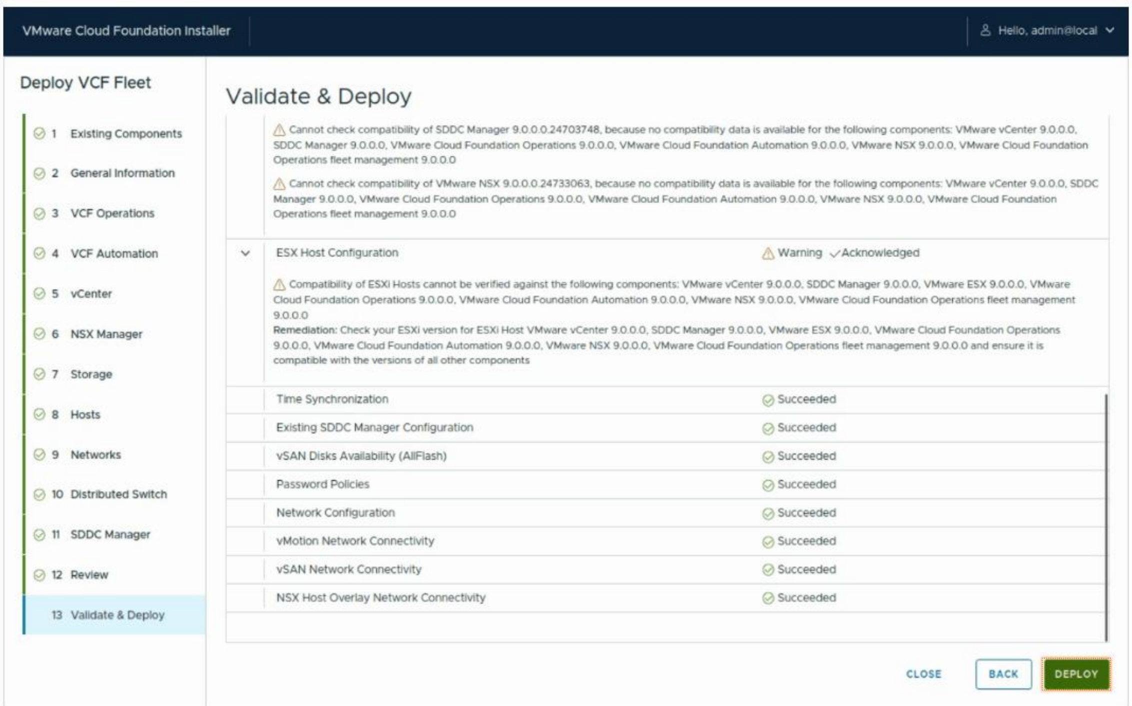


Validate & Deploy

VCF Installer will automatically validate the information entered.

Note: As this is a simulation lab, there will be a few benign warnings that can be safely ignored.

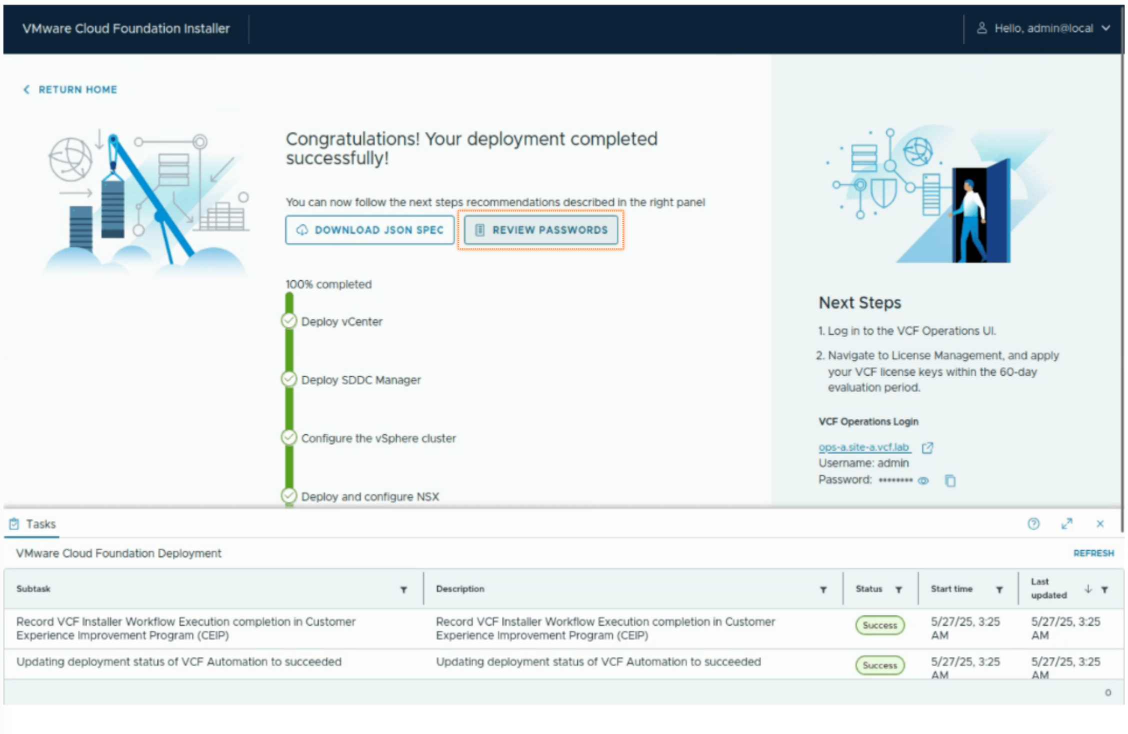
1. Click **ACKNOWLEDGE ALL WARNINGS** to ignore the warnings. All other validations should succeed with no errors.
2. Click on the vertical scroll bar to the right to scroll down the page to see more.
3. Click **DEPLOY** to proceed deploying VMware Cloud Foundation.



SUCCESS!

You have successfully deployed VMware Cloud Foundation. Let's review what had happened.

1. Click on the **Expand** arrows to view some of the tasks completed. The VCF Installer has successfully executed hundreds of tasks to deploy VCF automatically based on the inputs you had previously entered.
2. Click on the **Shrink** arrows to minimize the tasks view.
3. Click on the vertical scroll bar to the right to scroll up the page.
4. As we selected to auto-generate passwords, we can view the passwords generated for each of the components installed. Click **REVIEW PASSWORDS**.



The FQDN and credentials for each of the appliances installed can be viewed from this screen. The passwords can be changed from VCF Operations as a day 2 operation.

This concludes the VCF Installation simulation.